

UNIVERSITY OF EDINBURGH

INTERNSHIP REPORT

M1 INFORMATIQUE CONCEPTS ET APPLICATIONS

Varieties of Semiring Solvers via Multicategorical and Algebraic Constructions

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Introduction

- motivation → algebraic simplification is a pain to do manually in dependently typed languages/in proof assistants
- solvers in other languages (Rocq [GM05], Agda [Kid19])

read article a little bit

- state of the art: frex debut [YGK18], idris frex [All+25] and involution construction [All+23]
- presentation of the contributions:
 - finishing Ohad’s construction proof and correcting the needed hypothesis
 - proof of freeness of the construction
 - formalising the distributive combination theory in Idris
 - implementing the distributive combination semantics
 - implementing some free algebras
 - testing and comparison

1 Preliminaries

- universal algebra definitions (signatures, models, etc..., free algebras and free extensions)
 - finitary signature (with operators and arity)
 - algebra (with carrier and semantics)
 - terms
 - environment, interpretation
 - equation, valid equation
 - presentation (axioms)
 - model
 - model over a set
 - algebra homomorphism, model homomorphism
 - morphism of models
 - free model
 - extension
 - morphism of extensions
 - free extension
- cite [BS81], [All+25]
- definition: distributive combination of theories (**do not** write in terms of operads and translation, give a definition only involving presentations)

Do I put some category theory basics here ?

2 Categorical approach to the construction of distributive combinations

find better name for section

2.1 Free algebra for the distributive combination

- Challenge: not only constructing something that works, but something that is usable. Therefore, no use of quotients or pushouts or other hard to compute things. Can also not simply use the construction using coproducts described in [All+25], section 4.4.
- Definition of a commutative theory and an affine theory.
- recalling of what the free algebra gives us (unit and bind)
- **Theorem:**

Let $\mathcal{A}$ be a commutative theory, and $\mathcal{M}$ be an affine theory.

Suppose that we have $(F_{\mathcal{A}}, \eta^{\mathcal{A}}, \varepsilon^{\mathcal{A}})$ and $(F_{\mathcal{M}}, \eta^{\mathcal{M}}, \varepsilon^{\mathcal{M}})$ the free algebra constructions for $\mathcal{A}$ and $\mathcal{M}$.

We can construct the free algebra construction $(F_{\mathcal{M} \triangleright \mathcal{A}}, \eta^{\mathcal{M} \triangleright \mathcal{A}}, \varepsilon^{\mathcal{M} \triangleright \mathcal{A}})$ for the distributive combination $\mathcal{M} \triangleright \mathcal{A}$ using this data, and the construction is concrete.

reformulate

- Put the whole proof in the appendix (probably 50 pages long once all written out), but maybe explain what the key steps of the construction are

2.2 Relation with distributive laws

- define monads, algebras and distributive laws [Mac71]
- comparison with [ZM18]

2.3 An attempt at constructing the free extension for the distributive combination

- Proof that constructing the free extension for the combination from the two free extensions is possible.

finish this proof in my notes

- But proof not constructive (uses pushouts) $\rightarrow$ this is ok, this at least shows that it is theoretically possible.
- Failed proof under the same hypothesis as previous section $\rightarrow$ shows where the construction stops working: the construction lacks knowledge regarding the interaction of the concrete and abstract part of the free extension.

3 Implementation

3.1

name section

- maybe talk about setoids ? (and cite appropriate references)
- go over some basic definitions used such as `Signature`, `Op`, `Algebra`, `Term`, `SetoidAlgebra`, `Equation`, `Presentation`, `Model`
- explain the choice of representation for the theory of the distributive combination
- quickly go over the implementation of the semantics (the only real relevant part is `psi` but the definition in itself is atrocious, give type declarations only) → maybe give the definition of `DistributiveCombinationStructure`'
- don't go over the details of the axiom preservation and the freeness proofs (we don't have them anyway, also should we be straightforward about this ?)
- maybe only give the signature of `FreeDistributiveCombination`

3.2 Performance evaluation

- find interactivity and attention thresholds
- compare with the performances of the base solvers if possible

4 Comparison with existing solvers

find better name for section

only comparison with [GM05]

- quick description of how the `ring` tactic of Rocq is implemented (reflection)
- comparison of the theories supported → Rocq supports commutative semirings and commutative rings only, we support an umbrella of different combination of theories → put some sort of table, or the pseudo-matrix diagram of the first meeting

$$\underbrace{\begin{pmatrix} \text{commutative semigroup} \\ \text{commutative monoid} \\ \text{abelian group} \end{pmatrix}}_{\text{additive theory}} \times \underbrace{\begin{pmatrix} \text{semigroup} \\ \text{monoid} \\ \text{group} \end{pmatrix} \times \begin{pmatrix} \text{mere} \\ \text{commutative} \end{pmatrix} \times \begin{pmatrix} \text{mere} \\ \text{involutive} \\ \text{non-reversing involutive} \end{pmatrix}}_{\text{multiplicative theory}}$$

- pros of the tactic over the `fral`/`frex` approach (support for additional equations)
- comparison of capabilities (I believe that the `ring` tactic is not complete, whereas the `frex` approach is) → limitations of adding equations to the tactic, with the example of the involution

To check: I think that you can only add equations that use the know operations, i.e `+`, `*`, `0`, `1`.

- comparison in usability (runtime) → `frex` might be slower as typechecking takes a long time.

TODO: runtime comparison

Conclusion

References

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A Internship context

B Source code

Acknowledgments